Improving CMS Ratings & Business Performance: A Data-Driven Approach for Evanston Hospital

Healthcare Analytics Capstone Project

By

Ravi Kumar Thota

Shiva Shankar Iyer

Keshav Maheshwari

Amitabh Biswas

Executive Summary

Key Highlights:

- Evanston Hospital has been stuck at a **3-star CMS rating for 5 years**, impacting revenue and insurer preference
- CMS ratings are **outcome-driven**, not infrastructure-driven
- Machine learning analysis identifies **4 high-impact improvement areas**
- Targeted operational improvements can realistically move Evanston to **4 stars**

Business Problem Context

What’s going wrong?

- ✓ **7% decline in profits** over the last 6 quarters
- ✓ **5% annual decline** in patients and insurance partners
- ✓ Private healthcare demand is increasing, but **patients choose other hospitals**

Key Insight:

- ✓ Declining CMS ratings have directly reduced patient preference and insurer empanelment

Why CMS Ratings Matter?

CMS ratings directly impact:

- ✓ Patient trust & hospital selection
- ✓ Insurance network inclusion
- ✓ Long-term revenue sustainability

Challenge at Evanston:

- ✓ Investments made without understanding **which CMS measures matter most**
- ✓ For Evanston, a 3-star rating places it at a competitive disadvantage in a high-choice market

Root Cause Analysis

Phase 1

- **Issue Tree Framework**
  - To decompose the problem of stagnant profits and ratings

Phase 2

- **Data Understanding & EDA**
- To identify KPIs influencing ratings
- EDA: Perform univariate and bivariate analysis & feature engineering.

Phase 3

- **Model Building & Evaluation**
  - Hyperparameter optimization

Phase 4

- **Recommendations & Conclusion**
- Rating drivers and key insights

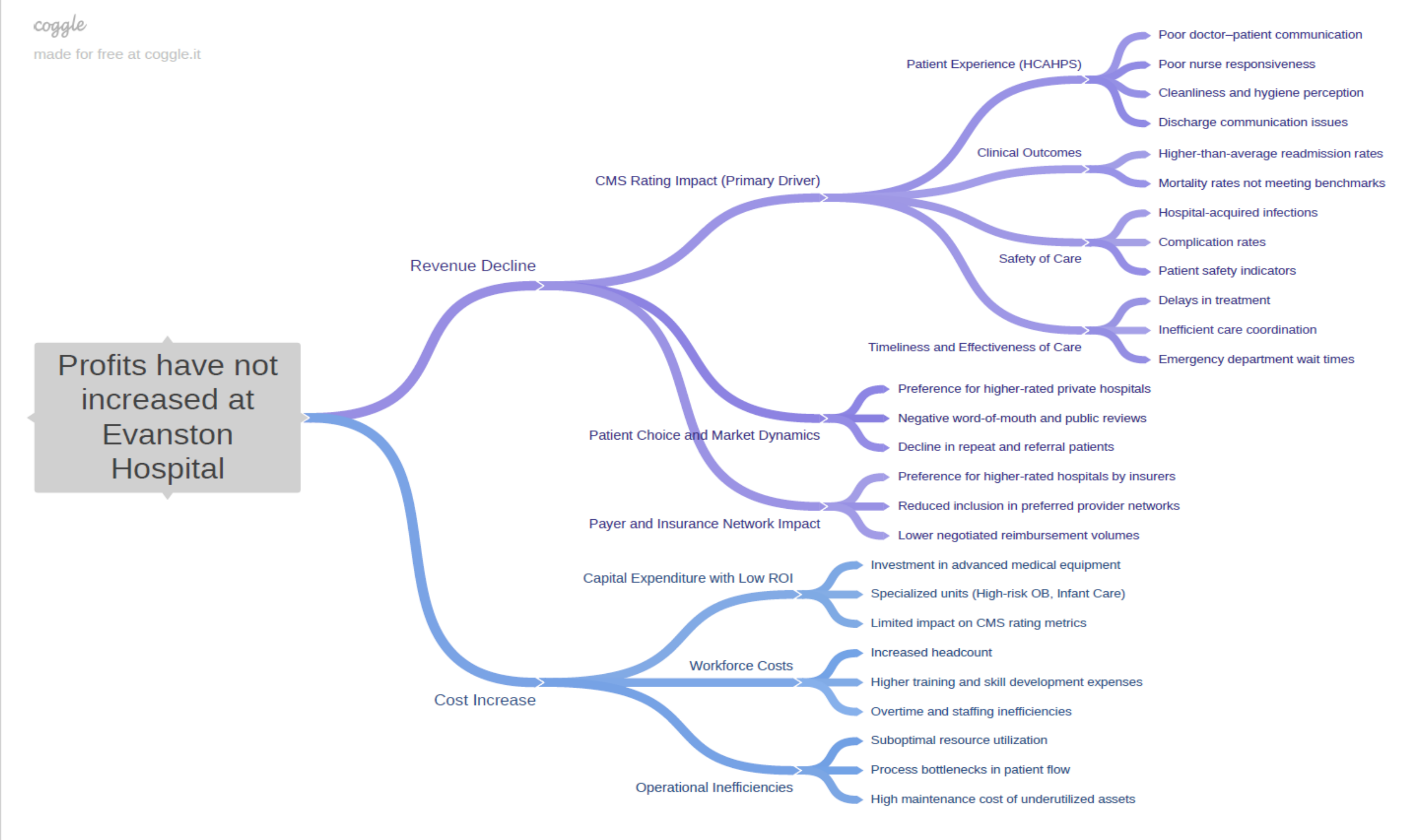
Issue Tree Framework

Key Root Causes Identified:

- ✓ Lack of understanding of **CMS rating methodology**
- ✓ Investments focused on **infrastructure over outcomes**
- ✓ No data-driven prioritization of improvement areas

Conclusion:

- ✓ Rating stagnation is a **decision-making problem**, not a capability problem



CMS Framework & Data Selection

CMS Rating Structure:

62 measures across **7 measure groups**

1. Mortality
2. Readmissions
3. Safety of Care
4. Timeliness
5. Effectiveness
6. Patient Experience
7. Imaging Efficiency

Data Source:
CMS Hospital Compare datasets

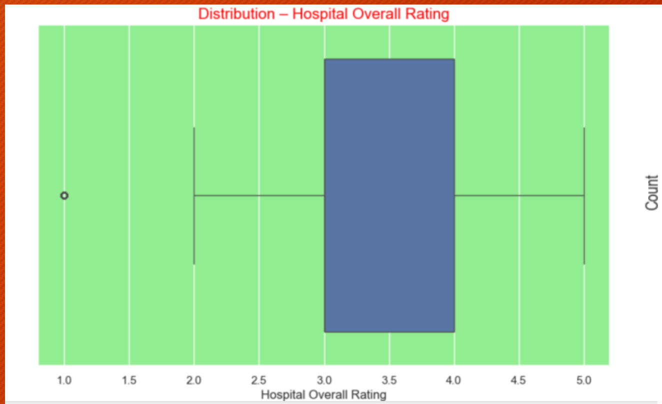
Data Preparation & EDA Highlights

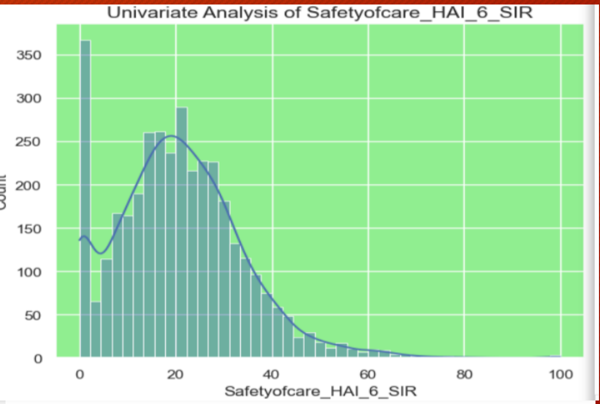
Key EDA Findings:

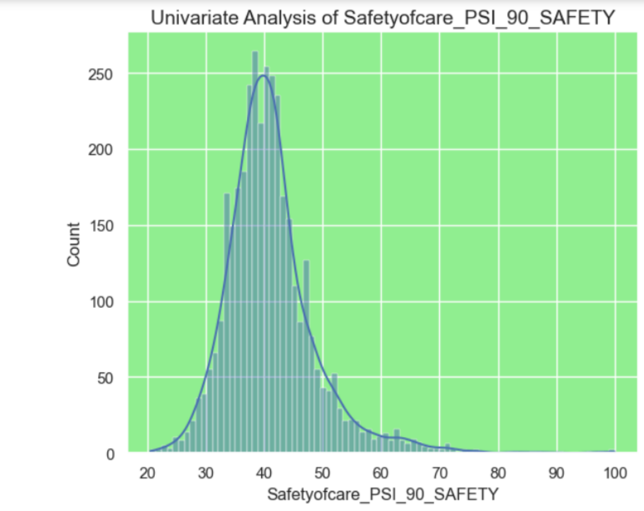
- Majority of hospitals cluster between **3-4 stars**
- Safety and Readmission metrics show **high variability**
- Certain measures exhibit skewness and required normalization

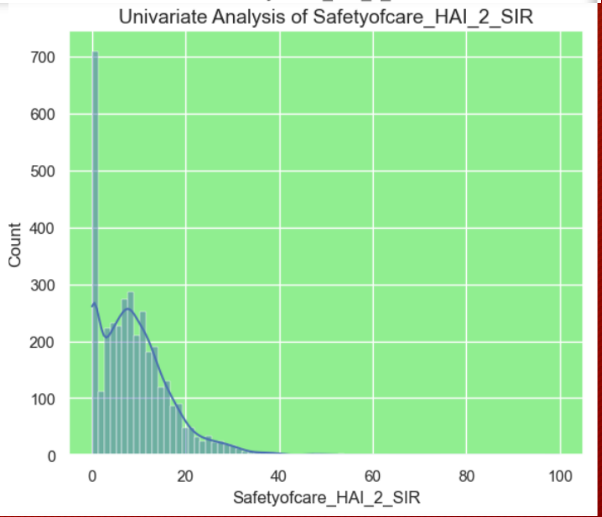
Insight:

- Small improvements in key metrics can **significantly shift ratings**









Modeling Strategy

Approach:

•Built classification models to predict CMS ratings

•Models tested:

- Baseline linear/logistic approach
- Logistic Regression with hypertuning
- Decision Tree Classifier
- Random Forest with hyperparameter tuning

Evaluation Focus:

•**Precision** → avoid misleading improvement focus
•**Accuracy** → balanced performance after CV

Model Performance

Top 10 Features for Evanston Hospital to Focus on:

1. **Patient Experience** - Clean Environment
2. **Readmission_COPD** - COPDisease - 30 days readmission rate
3. **Patient Experience** - Overall Rating of the Hospital Experience
4. **Mortality - Pnemonia** - 30 days mortality rate
5. **Readmission National Comparison** (Below national level)
6. **Safety of Care** - Catheter Associated Urinary Tract Infection
7. **Patient Experience** - Doctor Communication (H_COMP_2)
8. **Patient Experience National Comparison** (below & same)
9. **Mortality** - Acute Myocardial Infarction (AMI-Heart attack) mortality rate
10. **Mortality** - Heart Failure 30 days mortality rate

Weight	Feature
0.0958 ± 0.0573	PatientExperience_H_CLEAN
0.0633 ± 0.0365	Readmissions_READM_30_COPD
0.0536 ± 0.0823	PatientExperience_H_HSP_RATING
0.0485 ± 0.0442	Readmission national comparison_Below
0.0462 ± 0.0312	Mortality_MORT_30_PN
0.0397 ± 0.0299	Safetyofcare_HAI_2_SIR
0.0395 ± 0.0769	PatientExperience_H_COMP_2
0.0374 ± 0.0541	Patient experience national comparison_Same
0.0339 ± 0.0255	Mortality_MORT_30_HF
0.0330 ± 0.0618	Effectivenessofcare_MORT_30_AMI
0.0329 ± 0.0905	Patient experience national comparison_Below
0.0319 ± 0.0689	PatientExperience_H_QUIET_HSP
0.0307 ± 0.0683	PatientExperience_H_COMP_5
0.0294 ± 0.0255	Timelinessofcare_ED_2b
0.0285 ± 0.0430	Safety of care national comparison_Below
0.0258 ± 0.0275	Readmission national comparison_Same
0.0248 ± 0.0243	Timelinessofcare_OP_18b
0.0240 ± 0.0214	Medical_Imaging_OP_10
0.0237 ± 0.0201	Emergency Services
0.0234 ± 0.0202	Safetyofcare_PSI_90_SAFETY
	... 30 more ...

Model Performance Summary

Best Performing Model:

✅ Random Forest

Why Random Forest?

- ✓ Highest precision and accuracy
- ✓ Handles non-linear healthcare relationships
- ✓ Robust to noise across CMS measures

	Model Name	Accuracy	Precision	Recall	F1 Score
0	Linear Regression	80.0	-	-	-
1	Logistic Regression	73.9	77.8	59.5	65.16
2	Logistic Regression with hyper parameter tuning	72.77	75.11	54.74	60.06
3	Decision Tree Classifier	60.93	37.01	33.08	33.12
4	Random Forest With Hyperparameter Tuning	74.44	81.37	53.2	58.47

Outcome:

- ✓ Reliable identification of **high-impact improvement drivers**

Key Drivers Impacting Evanston’s Rating

Top Impact Areas Identified and Recommendations:

1. **Safety of Care -** Reduce hospital-acquired infections & complications
2. **Patient Experience -** Improve discharge communication and responsiveness
3. **Mortality Rates -** Strengthen high-risk clinical pathways
4. **Readmissions & Procedures -** Post-discharge follow-up and care coordination

Key Insight:

- **These factors influence ratings far more than imaging or infrastructure**
- From **capital expenditure** to **process excellence**
- Recommended execution priority: Safety → Experience → Mortality → Readmissions

Expected Impact

Expected Business Impact:

- Improved CMS rating → **higher patient inflow**
- Better insurer confidence → **network stability**
- Sustainable improvement → **long-term profitability**

“Data-driven prioritization—not guesswork—is the fastest path to a 4-star Evanston Hospital.”